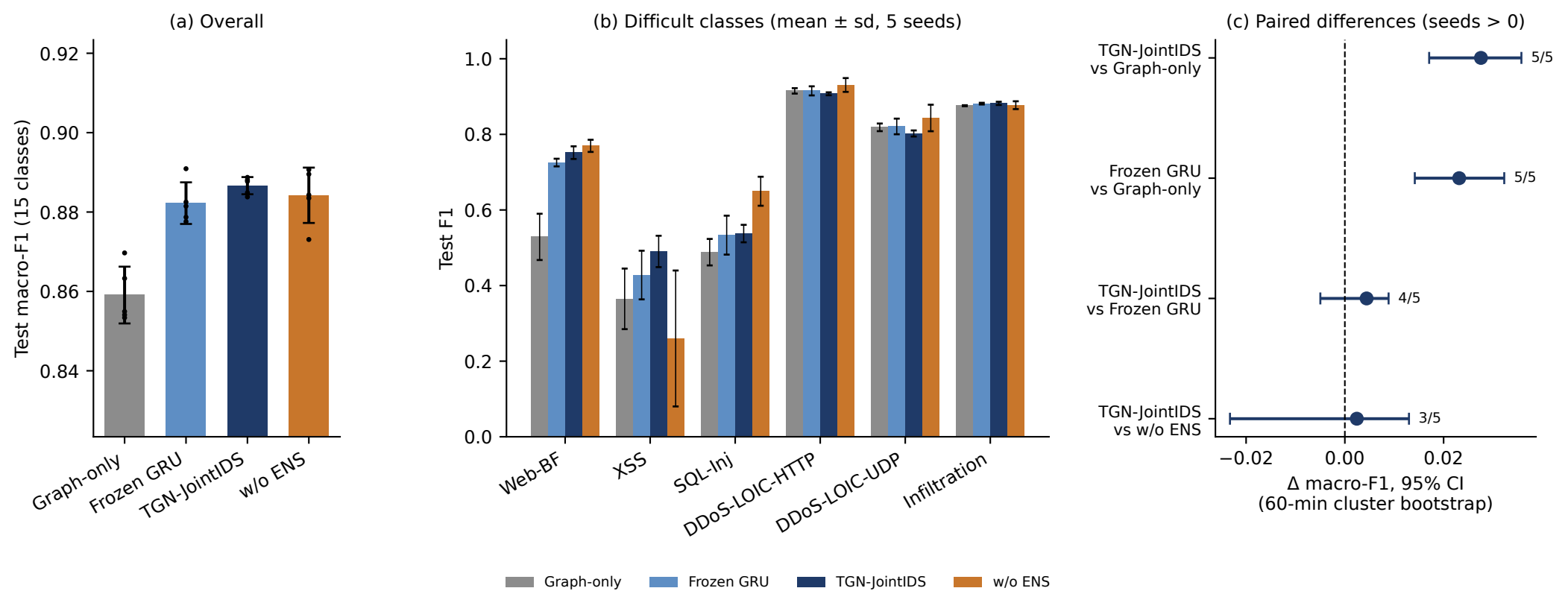
(a) Overall
Test macro-F1 (15 classes)
0.92
0.90
0.88
0.86
0.84
Graph-only
Frozen GRU
TGN-JointIDS
w/o ENS
(b) Difficult classes (mean ± sd, 5 seeds)
Test F1
1.0
0.8
0.6
0.4
0.2
0.0
Web-BF
XSS
SQL-Inj
DDoS-LOIC-HTTP
DDoS-LOIC-UDP
Infiltration
Graph-only
Frozen GRU
TGN-JointIDS
w/o ENS
(c) Paired differences (seeds > 0)
TGN-JointIDS vs Graph-only
5/5
Frozen GRU vs Graph-only
5/5
TGN-JointIDS vs Frozen GRU
4/5
TGN-JointIDS vs w/o ENS
3/5
−0.02
0.00
0.02
Δ macro-F1, 95% CI
(60-min cluster bootstrap)